coursera | UNIVERSITY OF LONDON

Mathematical Foundations for Computing

Module 1
Module 1: Sets and Set Theory

Module 2
Module 2: Number Systems and Bases

Lesson 2.1 Introduction to Number Bases
- Introduction to number bases
 Video • 13 min
- Introduction to number bases – conversion to decimal
 Practice Assignment • Grade: 100%
- Conversion of integer numbers between different bases
 Video • 7 min
- Conversion of non-integer numbers between bases
 Video • 14 min
- Non-integer numbers conversion
 Practice Assignment • Grade: 100%

Lesson 2.2 Operations with Binary Numbers
- Operations with binary numbers – addition
 Video • 3 min
- Operations with binary numbers – subtraction
 Video • 4 min
- Operations with binary numbers – multiplication
 Video • 4 min
- Number representations
 Video • 8 min
- Operations with binary numbers
 Practice Assignment • Grade: 100%
- Why do we need to use these number conversions?
 Dialogue • 15 min

Lesson 2.3 Summary and Assessments
- Topic 1 – looking back
 Video • 2 min
- Check your understanding: End of module 2
 Graded Assignment • Grade: 100%

Module 3
Module 3: Functions and Their Properties

Lesson 3.1 Introduction to functions
- The definition of a function
 Video • 11 min
- The definition of a function
 Practice Assignment • Grade: 100%
- Plotting functions
 Video • 9 min
- Plotting functions
 Practice Assignment • Grade: 100%
- Injective and surjective functions
 Video • 14 min
- **Injective and surjective functions**
 Reading • 15 min
- Injective and surjective functions
 Practice Assignment • 30 min
- 2D Graphing Calculator: Plotting some functions
 Lab • 20 min

Lesson 3.2 More about Functions
- Function composition
 Video • 4 min
- Function composition
 Practice Assignment • 30 min
- Bijective functions
 Video • 11 min
- Bijective functions
 Practice Assignment • 30 min
- Logarithmic functions
 Video • 12 min
- Logarithmic functions
 Practice Assignment • 30 min
- The floor and ceiling functions
 Video • 12 min
- The floor and ceiling functions
 Practice Assignment • 30 min
- Functions problem sheet and solutions
 Reading • 10 min

Lesson 3.3 Summary and Assessments
- Module 3 summary
 Reading • 15 min
- Check your understanding: End of module 3
 Graded Assignment • 20 min

Module 4
Module 4: Relations

Lesson 4.1 Understanding the concept of relations
- Definition of a relation: relation versus function
 Video • 8 min
- Definition of a relation: relation versus function
 Practice Assignment • 20 min
- Matrix and graph representations of a relation
 Video • 10 min
- Matrix and graph representations of a relation
 Practice Assignment • 20 min
- The properties of a relation: reflexive, symmetric and anti-symmetric
 Video • 14 min
- The properties of a relation: reflexive, symmetric and anti-symmetric
 Practice Assignment • 20 min
- Relation properties: transitivity
 Video • 5 min
- Relation properties: transitivity
 Practice Assignment • 20 min

Lesson 4.2 Equivalence, and partial and total order relations
- Equivalence relations and equivalence classes
 Video • 6 min
- Equivalence relations and equivalence classes
 Practice Assignment • 25 min
- Partial and total order
 Video • 6 min
- Partial and total order
 Practice Assignment • 30 min
- Building and analysing different types of mathematical relations
 Dialogue • 15 min

Lesson 4.3 Summary and Assessments
- Topic summary
 Reading • 15 min
- Check your understanding: End of module 4
 Graded Assignment • 20 min

Course Wrap-Up
- Mathematical Foundations for Computing: Course Summary
 Reading • 10 min
- The Archive Room: A Mathematical Escape Mystery
 Role Play • 15 min
- Course summary
 Video • 1 min

Injective and surjective functions

0:33 / 4:51 2x

Introduction to injective and surjective functions

Functions, also known as mappings or transformations, are fundamental concepts in mathematics. They describe relationships between sets. Two important types of functions are injective and surjective functions. Understanding these types of functions is crucial for studying various areas of mathematics, including algebra, calculus and discrete mathematics.

Injective functions

An **injective function** (or one-to-one function) is a function in which every element of the codomain is mapped by at most one element of the domain. In other words, no two different elements in the domain map to the same element in the codomain.

Definition

A function $f : A \to B$ is injective if $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for all $a_1, a_2 \in A$.

Graphical representation of a function

In a graphical representation of a function, if every horizontal line intersects the graph of the function at most once, the function is injective.

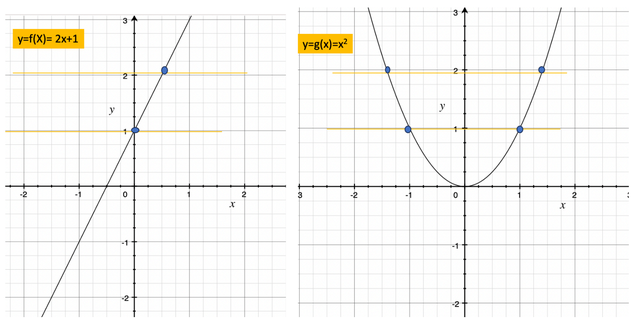


Image description: This figure shows the graphs for the representation of two functions: $f = 2x + 1$ on the left-side and $g = x^2$ on the right-side. The function f is an increasing straight line and is injective as all horizontal lines meet the curve of f at one point only. The function curve of the function g is decreasing for negative numbers and increasing for positive numbers. The graph shows that some horizontal lines intersect with the graph of g in more than one point. Therefore, the function g is not injective.

Example of an injective function

Consider the function $f : \{1, 2, 3\} \to \{a, b, c, d\}$ defined by $f(1) = a, f(2) = b, f(3) = c$. This function is injective because different elements in the domain map to different elements in the codomain.

Example of a non-injective function

The function $g : \{1, 2, 3\} \to \{a, b, c\}$ defined by $g(1) = a, g(2) = a, g(3) = b$ is not injective because both 1 and 2 map to the same element a in the codomain.

Surjective functions

A **surjective function** (or onto function) is a function in which every element of the codomain is mapped by at least one element of the domain. In other words, the function covers the entire codomain.

Definition

A function $f : A \to B$ is surjective if for every $b \in B$, there exists at least one $a \in A$ such that $f(a) = b$.

Graphical representation

In a graphical representation of a function, if every horizontal line intersects the graph of the function at least once, the function is surjective.

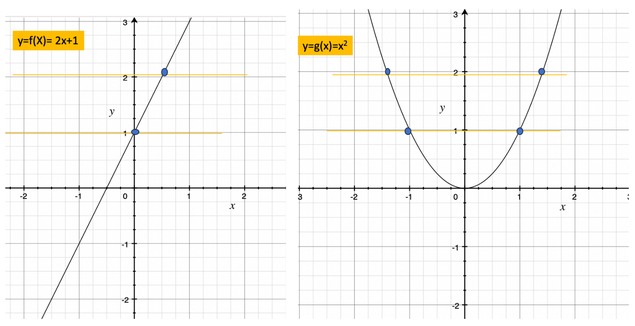


Image description: This figure shows the graphs for the representation of two functions: $f = 2x + 1$ on the left-side and $g = x^2$ on the right-side. The function f is an increasing straight line and is surjective as all horizontal lines meet the curve of f at least one point. The function curve of the function g is decreasing for negative numbers and increasing for positive numbers. The graphs show that all horizontal lines for which $y < 0$ don't intersect with the graph of g at any point. Therefore, the function g is not surjective.

Example of a surjective function

Consider the function $f : \{1, 2, 3\} \to \{a, b, c\}$ defined by $f(1) = a, f(2) = b, f(3) = c$. This function is surjective because every element in the codomain $\{a, b, c\}$ is mapped by some element in the domain.

Example of a non-surjective function

The function $g : \{1, 2\} \to \{a, b, c\}$ defined by $g(1) = a, g(2) = b$ is not surjective because c in the codomain is not mapped by any element in the domain.

Combining injective and surjective functions

A function can be both injective and surjective. Such functions are called bijective functions (or one-to-one correspondences).

Bijective functions

A function $f : A \to B$ is bijective if it is both injective and surjective. This means every element of the codomain is mapped by exactly one element of the domain.

Example

Consider the function $ f: \{1, 2, 3\} \to \{a, b, c\} $ defined by $ f(1) = a $, $ f(2) = b $, $f(3) = c $. This function is bijective because it is both injective and surjective.

Summary

- **Injective functions (one-to-one)**: No two different elements in the domain map to the same element in the codomain. Every element in the codomain is mapped by at most one element in the domain.
- **Surjective functions (onto)**: Every element in the codomain is mapped by at least one element in the domain.
- **Bijective functions (one-to-one correspondence)**: A function that is both injective and surjective. Every element in the codomain is mapped by exactly one element in the domain.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →